

# MH370 flaperon drift diagnostic - CMEMS GLORYS12 currents plus full WAVERYs surface Stokes

Uncalibrated family-specific diagnostic; currents=cmems\_glorys12\_reanalysis | isotope=enabled as a neutral-or-penalizing screen conditional on represented flaperon arrival

## Estimate at a glance

CURRENT FAMILY  
cmems\_glorys12\_reanalysis

PEAK SOURCE CELL  
arc-161

PEAK POSITION  
-34.900 deg , 93.117 deg

WEIGHTED TRAJECTORIES  
2048 x 77 cells

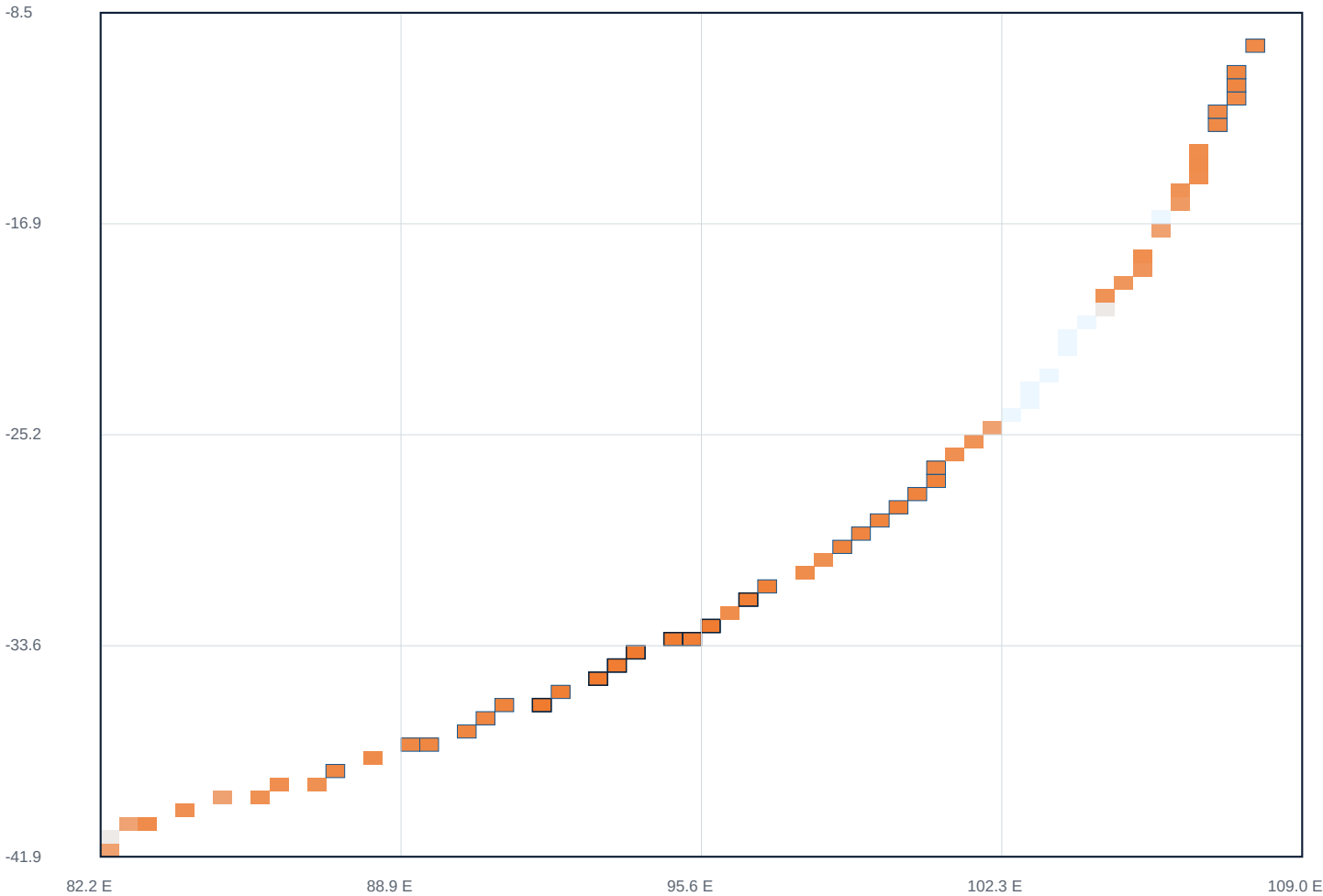
MEAN IMPORTANCE-WEIGHTED FIELD-TERMINATION FRACTION  
60.9%

MEAN PROPOSAL TERMINATION FRACTION  
39.6%

ARRIVAL SAMPLER  
AdaptiveSmc

PEAK CORRECTION ESS  
732.4

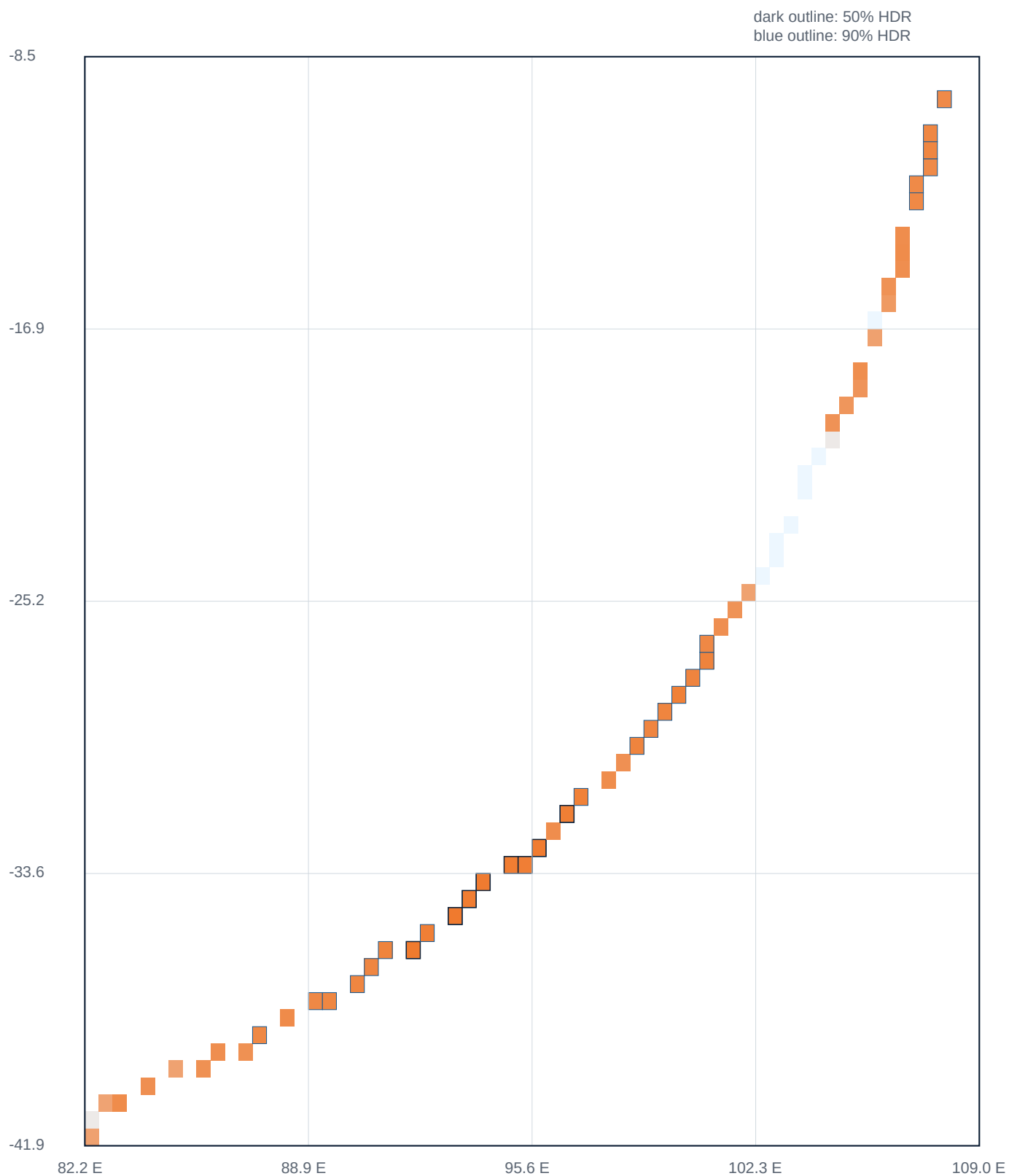
dark outline: 50% HDR  
blue outline: 90% HDR



Probability is conditional on the named motion, SATCOM, and evidence configuration.

# Posterior geography

Highest-density regions are computed from the weighted particle population.

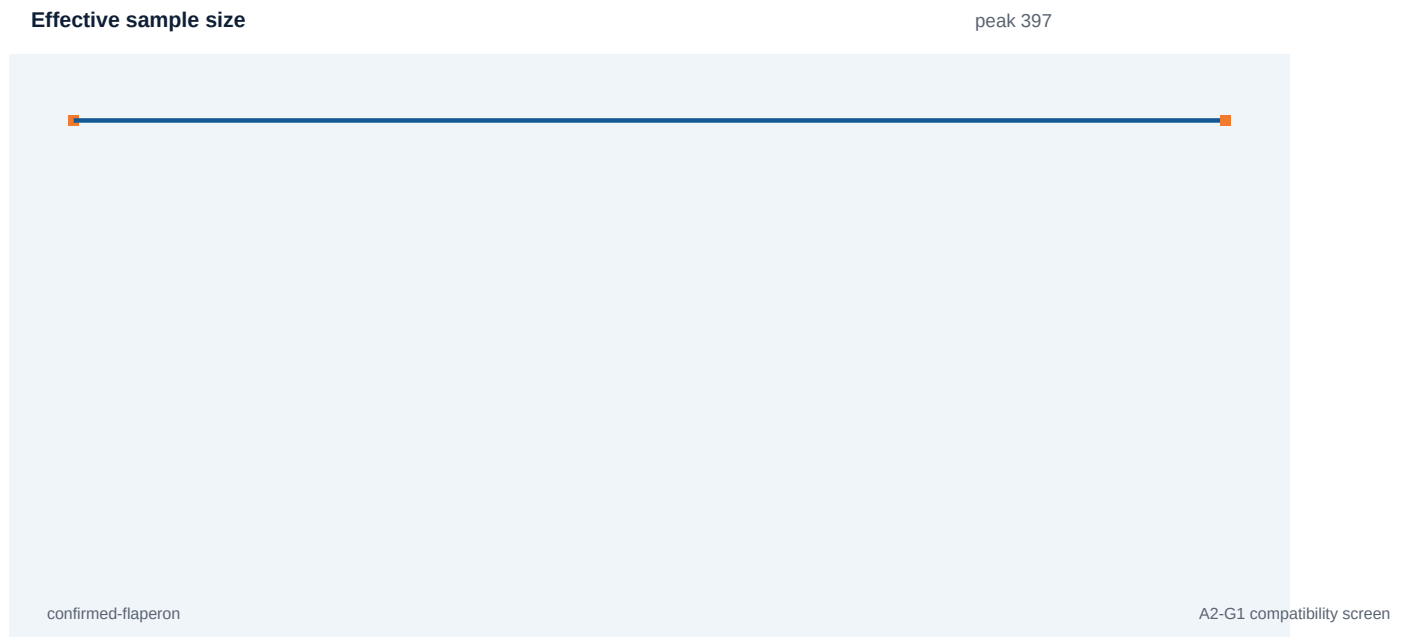


## Interpretation

The map is a probability density over the modelled impact state. It is not a searched-area elimination map, and structural model alternatives must be compared separately.

# Numerical behaviour and conditions

Adaptive likelihood tempering, particle diversity, and declared evidence conditions.



## Evidence conditions

- Current family selected explicitly: cmems\_glorys12\_reanalysis
- Current persistent-land corners renormalized=true; intermittent missing values remain fatal
- Barnacle isotope: enabled as a neutral-or-penalizing screen conditional on represented flaperon arrival
- direct windage disabled
- distinct Stokes velocity field enabled at 150% surface response; finite coastal corners renormalized=true
- explicit land mask enabled; coastal beaching disabled because no beaching-rate field was supplied
- Current interpolation across timestamp gaps greater than 36.0 h is outside-time support
- compatible=1448 marginal=7 rejected=1 missing\_SST=517

## Material limitations

- Diagnostic source-compatibility result, not a calibrated final MH370 posterior.
- Current families are reported separately and are never averaged using subjective weights.
- A2-G1 covers only the final months; the screen is neutral for any compatible chronology and never density-ranks compatible paths.
- No field-unavailable trajectory is relabelled as beaching; termination fractions and reasons remain in the JSON artifact.
- GLORYS12V1 is assimilative; the available drifter replay is not guaranteed held out.
- Daily means omit subdaily current variability represented by the native-HYCOM three-hour analyses.
- Land and any missing ocean value remain the packed missing sentinel; no missing value is converted to zero current.
- Stokes drift, direct windage, and coastal beaching are not included in this current field and require distinct explicit inputs.